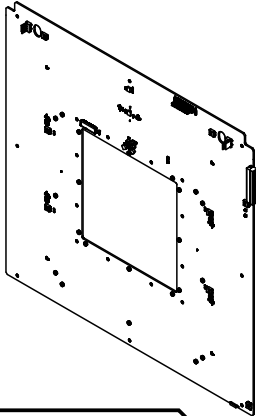
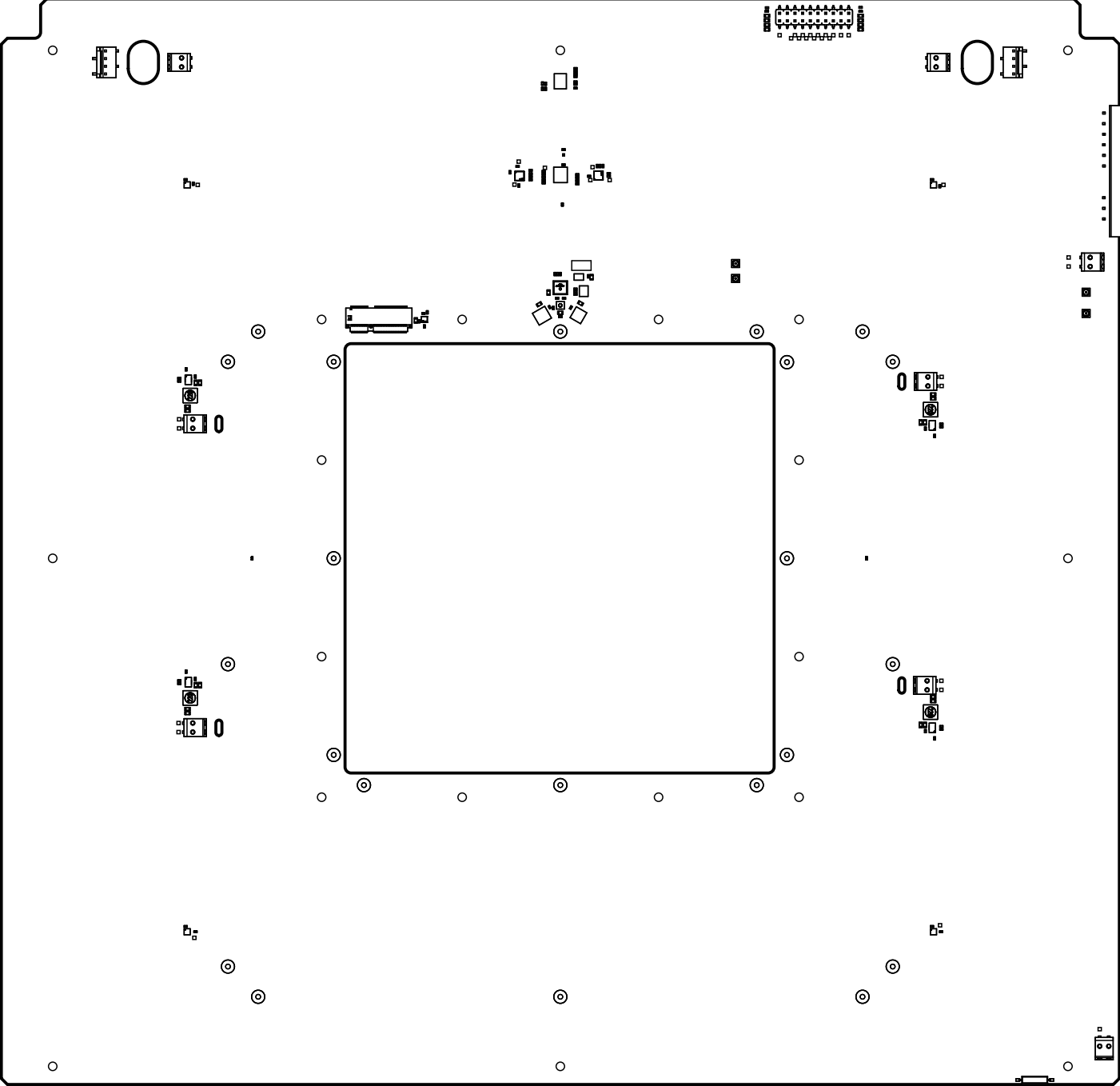


Assembly documentation:

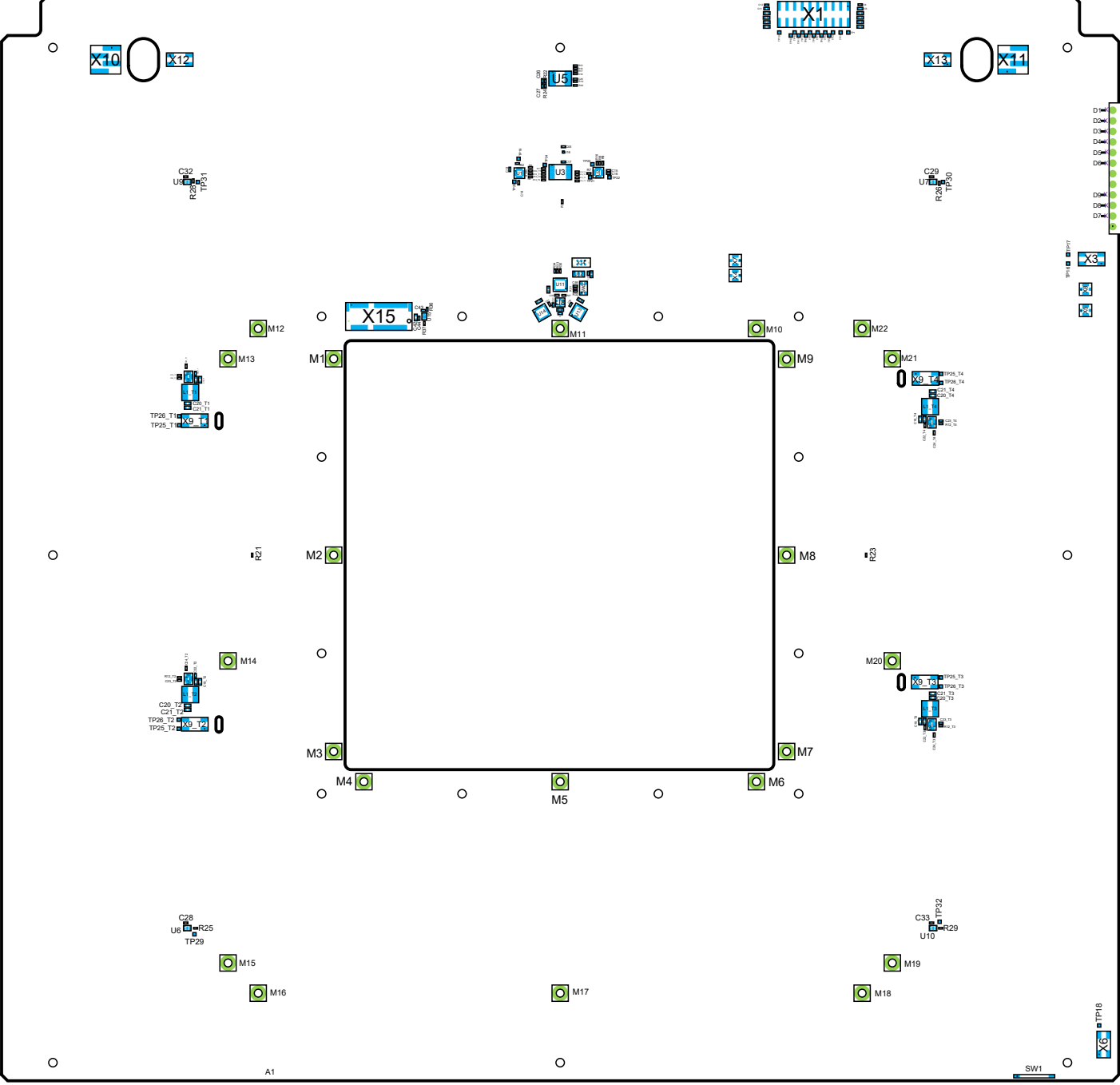
View from Top side (Scale 1:8)



View from Top side (Scale 1:2)



View from Top side (Scale 1:2)



- Notes:
- 1. Pin 1 marked with dot
 - 2. Cathode marked with K

ETH zürich			Project: STB Sensor Board	
Drawing number:	Rev: A	Format: A3	Laboratory: PBL	Sheet: STB_SensorBoard_assembly.PCBDwf
Date:	10.11.2022		Drawn by: gruprecht	Page: 1 of 2
File: C:\Users\phili\polybox\05_AltiumProjects\SolarTestBed\STB_SensorBoard\STB_SensorBoard_assembly.PCBDwf				

Bill Of Materials

Line #	Quantity	Designator	Name	Value	Tolerance	Power	Footprint	Manufacturer	Manufacturer Part Number
	25	C1, C2, C3, C4, C7, C8, C9, C10, C18_T1, C18_T2, C18_T3, C18_T4, C19_T1, C19_T2, C19_T3, C19_T4, C20_T1, C20_T2, C20_T3, C20_T4, C21_T1, C21_T2, C21_T3, C21_T4, C42	10uF	10uF	±10%		CAPC0603(1608)100_L	Samsung	CL10A106MP8NNNC
	2	C5, C11	1nF	1 nF	10%		CAPC1005X55X25ML5T10	Murata	GRM155R71H102KA01D
	33	C6, C12, C13, C14, C15, C16, C17, C22_T1, C22_T2, C22_T3, C22_T4, C23_T1, C23_T2, C23_T3, C23_T4, C24_T1, C24_T2, C24_T3, C24_T4, C25, C28, C29, C31, C32, C33, C35, C37, C40, C41, C43, C44, C45, C46	100nF	100nF			CAPC1050X55X23LL05T13	Samsung Electro-Mechanics	CL05B104KO5NNNC
	2	C26, C27	100pF	100pF			CAPC1005X56X25LL10T15	AVX Interconnect / Elco	04023A101KAT2A
	5	C30, C34, C36, C38, C39	1uF	1uF			CAPC1608X90X35LL15T15	Kemet	C0603C105K4RACTU
	7	D1, D2, D3, D4, D5, D6, D7	ESD9R3.3ST5G				ONSC-SOD-923-2-514AB-01_V	On Semiconductor	SZESD9R3.3ST5G
	1	D8	ESD9C5.0ST5G				ONSC-SOD-923-2-514AB-01_V	On Semiconductor	ESD9C5.0ST5G
	1	D9	ESD9X12ST5G				ONSC-SOD-923-2-514AA-01_V	On Semiconductor	ESD9X12ST5G
	4	L1_T1, L1_T2, L1_T3, L1_T4	744042001	1 µH	30%		WE-TPC-4818-M	Würth Electronics	744042001
	1	L2	LI0402E300R-10				INDC1005X07L	Laird-Signal Integrity Products	LI0402E300R-10
	22	M1, M2, M3, M4, M5, M6, M7, M8, M9, M10, M11, M12, M13, M14, M15, M16, M17, M18, M19, M20, M21, M22	9774015243R				9774015243R	Würth Electronics	9774015243R
	28	R1, R2, R7, R11, R12_T1, R12_T2, R12_T3, R12_T4, R13_T1, R13_T2, R13_T3, R13_T4, R14_T1, R14_T2, R14_T3, R14_T4, R19, R20, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34	10k, DNP	10 kOhm	1%		RESC1005X40X25LL05T05	Panasonic	ERJ-2RKF1002X
	6	R3, R4, R6, R9, R10, R36	100k	100 kOhm	1%		RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402FR-07100KL
	1	R5	200k	200 kOhm	1%		RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402FR-07200KL
	1	R8	560k	560 kOhm	1%		RESC1005X40X25LL05T05	Panasonic	ERJ-2RKF5603X
	4	R15, R16, R17, R18	4k7	4.7 kOhm	1%		RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402FR-074K7L
	2	R21, R23	NCP15XH103F03RC	10 kOhm	1%		FP-NCP15-NCU15-0_05-IPC_C	Murata	NCP15XH103F03RC
	2	R22, R24	1k6	1.6 kOhm	1%		RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402FR-071K6L
	1	R37	1M	1 MOhm	1%		RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402FR-071ML
	1	SW1	MK17-B-2				SMEL-MK17-B-2_V	Standex Meder	MK17-B-2
	1	SW2	DNP				PNSC-EVPAA_V	Panasonic	EVP-AA002K
	2	U1, U2	AD5593RBCPZ-RL7				FP-CP-16-32-IPC_C	Analog Devices	AD5593RBCPZ-RL7
	1	U3	PCA9646PW,118				TSSOP-16_N	NXP Semiconductors	PCA9646PW,118
	4	U4_T1, U4_T2, U4_T3, U4_T4	MP8833GD-0000-Z				QFN40P200X300X100-16N	Monolithic Power Systems	MP8833GD-0000-Z
	1	U5	MAX6615AEE+T				MAXM-E16+1_M	Maxim	MAX6615AEE+T
	4	U6, U7, U9, U10	OPT3001DNPT				DNP0006A	Texas Instruments	OPT3001DNPT
	1	U8	VEML60311X00				6-SMD - No Thermal Pad_2	Vishay	VEML60311X00
	1	U11	AS72651-BLGT				FP-AS72651-BLGT-MFG	ams	AS72651-BLGT
	1	U12	MX25L4006EZUI-12G				MXIC-USON-8_V	Macronix	MX25L4006EZUI-12G
	1	U13	AS72653-BLGT				FP-AS72653-BLGT-MFG	ams	AS72653-BLGT
	1	U14	AS72652-BLGT				FP-AS72652-BLGT-MFG	ams	AS72652-BLGT
	1	U15	TPS25200DRVT				FP-DRV0006A-MFG	Texas Instruments	TPS25200DRVT
	1	U16	N24S64BC4DYT3G				4-WLCSP (0.77x0.77)	ON Semiconductor	N24S64BC4DYT3G
	1	X1	61002021121				61002021121	Würth Elektronik	61002021121
	1	X2	1844317				1844317	Phoenix Contact	1844317
	8	X3, X6, X9_T1, X9_T2, X9_T3, X9_T4, X12, X13	2059-302/998-403				2059-302/998-403	WAGO	2059-302/998-403
	4	X4, X5, X7, X8	73412-0110				MOLX-73412-0110_V	Molex	73412-0110
	2	X10, X11	3-647168-4				3-647168-4	TE Connectivity	3-647166-4
	1	X15	2199119-4				2199119-4	TE Connectivity	2199230-4



Project:

STB Sensor Board

Drawing number:

Date: 10.11.2022

Rev: A

Format:
A3

Laboratory: PBL

Drawn by: gruprecht

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